Progress Report

Prompt-Driven Interactive Segmentation as a Dataset-Annotation Tool

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Milestone 2 — November 1, 2026
CSE 681 — Digital Image Processing

Abstract

This project investigates whether point- and box-prompted SAM 2.1 can reduce the effort required to produce instance masks while retaining agreement with COCO ground truth. The local dataset workflow now contains 300 deterministically selected COCO 2017 validation images, 979 non-crowd instances across 80 categories, and exported instance masks. The React/TypeScript interface and FastAPI backend now support browsing those images, selecting a target instance, and saving a prediction with its annotation time and computed IoU/Dice. The software checks pass, and one real FastAPI segmentation request has generated a SAM 2 mask on the local MPS device. This smoke test is not a quality or time-savings result: the manual baseline remains unmeasured, and no SAM 2 mask has yet been scored against ground truth.

1 Recap

Manual instance-mask annotation is labor-intensive. We compare a manual polygon baseline with a human-in-the-loop workflow in which an annotator prompts a frozen SAM 2.1 Hiera-Tiny model and accepts or refines its mask. Both methods are intended to annotate the same COCO object instances. The data source is COCO 2017 val2017 (Lin et al., 2014); the reproducible subset contains 300 images selected with seed 42 from images with 1–8 non-crowd instances. The selected annotations yield 979 non-crowd instance masks in 80 categories. The reference mask is intended for scoring only, not for choosing prompts.

2 Implementation Status

The current application uses a React, TypeScript, and Vite frontend with a Python FastAPI backend. The backend exposes local COCO image listing and image serving, accepts prompted image-segmentation requests, and provides an annotation-saving endpoint that checks image/annotation pairing, computes binary-mask IoU and Dice, and records the mask, prompts, and elapsed time in a local evaluation manifest. The frontend includes COCO image and target-instance selectors, positive/negative point and box prompts, mask review, and acceptance with metric display. Downloaded images, masks, checkpoint, and evaluation outputs are excluded from Git; reproduction instructions are tracked. The images retain their individual terms of use (COCO Consortium).

The COCO data/API tests pass (17 backend tests in total), and the frontend TypeScript build and lint checks pass. These are software-validation results, not segmentation experiments. The optional SAM 2 dependency and downloaded checkpoint were configured together, and the

FastAPI segmentation route returned a mask for one local COCO image using Apple MPS. The original plan's manual polygon baseline and multi-instance evaluation remain to be run. No model training is planned: the contribution is an interactive annotation and evaluation workflow around a frozen SAM 2 model (Ravi et al., 2024).

2.1 Baseline Reproduction

The manual polygon baseline has not yet been reproduced or measured. No annotation times, IoU, or Dice values are available for it. It remains a required M3 experiment: annotate the same target objects with a standard polygon tool, using the same timer start/stop definition, then compare masks with the corresponding COCO references.

2.2 Your Method

The web workflow is implemented through the software path: browse a local COCO sample, choose one of its non-crowd objects, add points and/or a box, request a mask, and accept it. Acceptance saves the prediction and elapsed time and computes metrics against the matching exported COCO mask. Inference latency is reported separately from annotation time. One call to `/api/segment` with a positive point prompt returned a 428×640 mask with model confidence 0.954. The prompt was placed by visual inspection, without using the reference mask. This confirms the configured inference path but is not a ground-truth quality score or a completed annotation; saving and scoring real accepted annotations remains necessary.

3 Preliminary Results

The current quantitative evidence describes dataset preparation, software validation, and one unscored inference smoke test. No human-created baseline masks or SAM 2 predictions have been evaluated, so method-level IoU, Dice, and annotation-time comparisons are not yet available.

Table 1: Evidence available at the time of this progress report. "Not measured" denotes missing experimental data, not a zero score.

Workflow	**Samples**	**Mean IoU / Dice**	**Observed time**
Manual polygon baseline	0 measured	Not measured	Not measured
SAM 2.1 smoke test	1 generated; 0 scored	Not measured	1.56 s per first API request
Prepared COCO subset	979 masks	–	–
Automated backend tests	17 passed	–	–

The smoke-test request used the 428-by-640 COCO image `000000000776.jpg` and one positive point at $(130, 130)$, selected by visual inspection. It ran on Apple MPS with the SAM 2.1 Hiera-Tiny checkpoint. The 1.56-second request includes predictor initialization and must not be interpreted as steady-state latency or human annotation time. The returned model confidence was 0.954; it is not an IoU/Dice measurement. No representative success/failure gallery is included because the generated mask has not yet been qualitatively reviewed.

4 Challenges & Deviations from the M1 Plan

The local subset and mask export are ready, and the model/checkpoint now produce a real mask through the API. However, installation and a one-image smoke test do not establish annotation quality or time savings. The manual baseline and paired evaluation remain behind the software implementation. M2 therefore prioritized a reproducible data/API path and persisted metrics; experimental validation is still incomplete. COCO source images and derived masks remain local and ignored by Git; the tracked data guide describes how to reproduce them.

5 Updated Work Breakdown & MVP Status

The software portion of the MVP is implemented, but the M1 minimum viable experimental deliverable has not been reached: the fixed-sample manual and prompted masks, paired timing data, and comparative metrics are still missing. The team must complete these measurements before drawing conclusions in M3. Because the team has two members, Student B combines the planned model/tool and evaluation/integration responsibilities.

Table 2: Updated two-member milestone allocation and status.

Milestone	**Student A**	**Student B**
M2 (current)	COCO subset and manual-baseline protocol prepared; polygon measurements remain pending.	React/FastAPI workflow implemented and one SAM 2 inference smoke test completed; evaluated samples remain pending.
M3 (planned)	Complete the manual polygon baseline on the fixed sample and record time consistently.	Install and validate SAM 2.1, complete prompted annotations, audit metrics, and produce the comparison and qualitative examples.

6 Next Steps

1. Review real masks from point- and box-prompt runs and verify the accept-and-save path against the matching ground-truth masks.

2. Select and record the fixed evaluation sample and target annotation IDs before either method is run; do not reveal ground truth during prompt choice.

3. Collect paired manual polygon and prompted masks and times using the same timer definition. Record device, model/checkpoint identity, prompts, failures, and any excluded cases.

4. Run the evaluator, verify per-instance mask dimensions and IDs, and report mean/median annotation time, mean IoU and Dice, and representative successes and failures. Avoid time-savings claims until these measurements are complete.

References

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